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RESEARCH ARTICLE

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From social media to flood 3D scenes: leveraging large language models and knowledge graphs for flood spatiotemporal evolution representation

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ABSTRACT

Flood disaster evolution exhibits strong spatiotemporal coupling and complex causality. Intuitive representations of flood disaster evolution can help the public better understand disaster processes and their underlying evolution mechanisms, thereby enhancing risk awareness. However, most existing studies focus on numerical modeling of flood disaster evolution, which has limited capability in effectively capturing and representing the evolving processes of flood disasters. Social media posts generated during floods contain rich information and provide valuable insights into the evolution of flood events. In this context, this study leverages large language models (LLMs) and knowledge graphs (KGs) to extract flood spatiotemporal knowledge from social media, thereby reconstructing the disaster evolution pathway for representation as 3D scenes. Specifically, ontology-constrained LLMs for flood knowledge extraction, evolutionary pathway reconstruction and 3D representation are introduced in detail. We collected a total of 34,731 tweets related to the 2021 European floods for experimental analysis and developed a prototype system for 3D flood representation. The experimental results demonstrate that the proposed method achieves reliable performance in disaster knowledge extraction and evolutionary pathway reconstruction. Moreover, the augmented 3D representation improves the interpretability and clarity of flood disaster evolution, thereby facilitating a better understanding of disaster processes and supporting disaster prevention and mitigation.

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Flood spatiotemporal evolution; social media; knowledge graphs; large language models; 3D representation

1. Introduction

Flood disasters, characterized by sudden onset, wide spatial extent and severe impacts, pose persistent threats to global societal safety and sustainable development. The United Nations Office for Disaster Risk Reduction (UNDRR[1]) reports that floods are the most frequent and widely affecting natural disasters worldwide (UNDRR 2020). According to the report of the European Environment Agency (EEA), floods accounted

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for more than 40% of the economic losses caused by weather and climate-related extreme events in Europe between 1980 and 2022. Particularly, the extreme floods of July 2021 caused widespread damage in Germany, Belgium, the Netherlands and neighboring countries, ranking among the most severe basin-scale flood disasters in Europe in recent decades (European Environment Agency 2023). Under ongoing climate change, the increasing occurrence of extreme rainfall has further intensified the uncertainty and complexity of flood risk (Mantovani *et al.* 2024). Enhancing flood resilience through modern information technologies has therefore become a critical priority for global disaster risk reduction (Peng *et al.* 2022).

Understanding flood disaster evolution is essential for leveraging information technologies to support disaster prevention and mitigation (Saunders *et al.* 2025). Flood disasters are inherently dynamic processes characterized by strong spatiotemporal coupling and interactions among multiple events. Their evolution can be described as a sequence of interconnected stages involving onset, propagation, emergency response and secondary impacts (Zhu *et al.* 2022). Therefore, accurate characterization of flood spatiotemporal evolution and key events provides a scientific basis for flood prevention and management (Gao *et al.* 2025, Chen *et al.* 2026). However, most existing studies rely on hydrodynamic models to simulate flood processes and integrate the results into GIS platforms for visualization (Kumar *et al.* 2023). Although these approaches can describe physical processes in detail, they have limited ability to coherently represent semantic relationships and event evolution structures, which constrains a comprehensive understanding of flood spatiotemporal evolution (Hofmeister *et al.* 2024).

Within networked information environments represented by social media, large volumes of user-generated data provide new insights to characterize the evolution of flood disasters. Compared with traditional monitoring methods, social media data offer high timeliness, broad coverage and strong public participation. During flood events, large numbers of users continuously publish disaster textual information. Although individual posts generally reflect only local phenomena or personal perceptions, the continuous accumulation of such fragmented information across temporal and spatial dimensions can support the identification of relationships among disaster events and the reconstruction of disaster evolution processes (Karmegam *et al.* 2021, Li *et al.* 2023). However, this information is typically scattered across large volumes of text in a fragmented and unstructured form, and therefore cannot be readily used to support systematic analysis or an integrated representation of flood disaster evolution (Phengsuwan *et al.* 2021). This limitation highlights the need for methods that can effectively organize and semantically connect dispersed disaster-related information from social media.

Knowledge graphs (KGs) are widely used to organize and represent complex entities, events, and their relationships, providing an effective means to structure and associate large volumes of disaster information (Kejriwal 2022, Peng *et al.* 2023). In recent years, a new generation of large language models (LLMs), represented by ChatGPT, Gemini, DeepSeek and Qwen, has demonstrated strong capabilities in natural language understanding, contextual modeling and complex reasoning. These capabilities provide new opportunities for automatically extracting disaster entities, relations

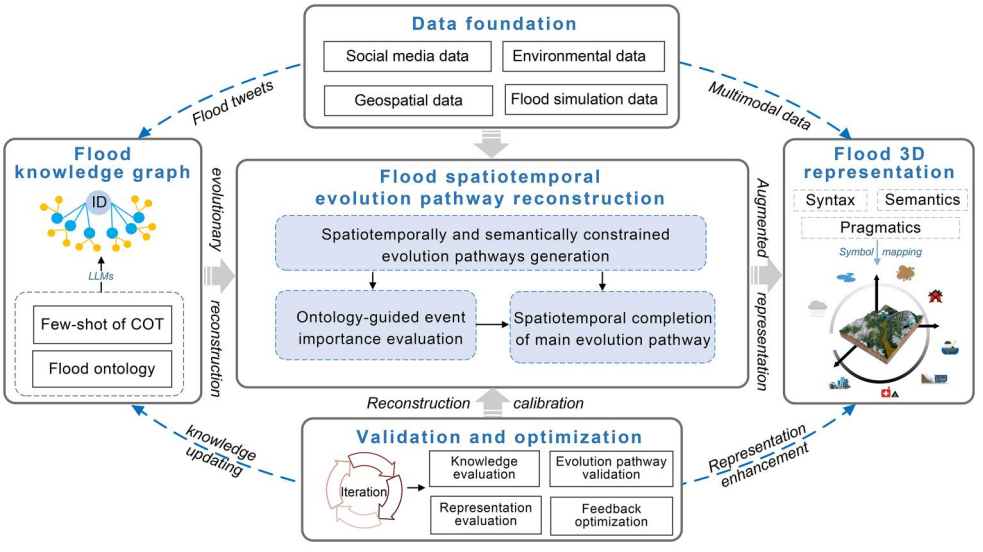


Figure 1. Overall framework for flood spatiotemporal evolution representation from social media. The framework integrates multimodal data, ontology-constrained flood knowledge extraction, flood spatiotemporal evolution pathway reconstruction, 3D flood representation and validation–optimization feedback. Social media data are used to construct an evidence-grounded flood knowledge graph through few-shot CoT prompting and FO constraints. Based on the graph, spatiotemporal and semantic constraints, ontology-guided event importance evaluation, and spatiotemporal completion are applied to reconstruct flood evolution pathway, which is further mapped into dynamic 3D geographic scenes.

and events from unstructured social media texts (Xue *et al.* 2023, Dang *et al.* 2025, Xu *et al.* 2025). Therefore, integrating LLMs with KGs can enable the structured organization of disaster knowledge from fragmented social media information and further support relational analysis and reconstruction of flood disaster evolution. However, node- and relation-centered graph representations alone remain insufficient to support intuitive public understanding of flood disaster evolution (Zajac *et al.* 2025). Geovisualization is widely recognized as an effective approach for understanding the evolution of complex geographic phenomena. Through spatial mapping and dynamic representation, abstract semantic logic can be transformed into perceptible spatiotemporal evolution processes (Guo *et al.* 2005, Lin *et al.* 2020, Bae and Dodge 2023).

In this context, this study leverages LLMs and KGs to extract flood spatiotemporal knowledge from social media, reconstruct the flood disaster evolution pathway, and present it in the form of 3D scenes. The study workflow is shown in Figure 1. This method exploits the semantic understanding and reasoning capabilities of LLMs to extract entities and relationships from large volumes of flood tweets, thereby constructing a flood KG. Based on this graph, the evolution pathway of flood is reconstructed under spatiotemporal and semantic constraints, and the corresponding entities and relationships are instantiated to generate 3D flood scenes, which in turn enhance public understanding of the development of floods. The main contributions of this study are summarized as follows.

1.1. A geographic scene knowledge-constrained LLM approach for flood knowledge extraction and spatiotemporal evolution pathway reconstruction

Taking flood tweets as the basis, this study follows the concept of the six elements of geographic scenes, namely time, place, character, object, event and phenomenon, which are modeled as an ontology and injected into LLMs as prior knowledge to constrain hallucinations. By adopting different prompting strategies, including zero-shot prompting, few-shot Chain-of-Thought (CoT) prompting and few-shot CoT prompting with Flood Ontology (FO) constraints, the proposed approach improves the quality of structured flood knowledge extraction. Furthermore, the extracted knowledge is used to reconstruct the spatiotemporal evolution pathway of flood disaster under temporal, spatial and semantic constraints, which serve as the storyline for subsequent 3D dynamic representation of flood disaster evolution.

1.2. A knowledge-guided 3D dynamic representation method for flood spatiotemporal evolution

Following the above-mentioned spatiotemporal evolution pathway, this study further maps flood event nodes and relations into 3D geographic scenes from a geospatial storytelling perspective that integrates the syntax–semantics–pragmatics framework, enabling a full-process representation from social media information to 3D flood scenes. The proposed method intuitively presents the occurrence, development, impacts and evolutionary logic of flood disasters, thereby enhancing the interpretability of flood disaster evolution, improving the effectiveness of flood information dissemination, and potentially strengthening public flood risk awareness.

The remainder of this article is organized as follows. Section 2 reviews related work. Section 3 presents the methodology. Section 4 describes the experiments and results. Section 5 discusses the findings, limitations and future directions. Section 6 concludes the article.

2. Related work

Social media has become an important data source for flood disaster perception and analysis. Existing studies have used platforms such as Twitter for disaster topic identification and public sentiment analysis during flood events (Bryan-Smith *et al.* 2023, Li *et al.* 2023). Topic modeling and feature extraction methods have also been applied to monitor disaster impacts from social media streams (Mihunov *et al.* 2022). Beyond textual semantic analysis, Twitter information has been used for rapid flood mapping and for analyzing disaster resilience and public responses (Li *et al.* 2018, Zou *et al.* 2018). Furthermore, event modeling of social media data has been employed to describe flood development from prewarning through emergency response to post-disaster recovery (Liu *et al.* 2022). These studies demonstrate that social media data can not only reflect local disaster conditions and public perceptions but also provide valuable support for analyzing flood disaster evolution. However, conventional topic models, feature engineering approaches and event modeling methods often rely on predefined rules or task-specific models, limiting their ability to adequately handle

noisy expressions, abbreviations, non-standard place names, missing context and implicit semantic relationships in social media texts (Wang *et al.* 2020, Chen and Lim 2021).

With the development of GeoAI, LLMs have increasingly been applied to geographic text understanding and disaster knowledge acquisition (Hu *et al.* 2023, Yang *et al.* 2026, Zhao *et al.* 2026). Compared with traditional rule matching methods (Li *et al.* 2022a), LLMs provide stronger contextual modeling and semantic understanding capabilities, making them better suited to handling noisy expressions, abbreviations, non-standard place names and missing context in social media texts. Nevertheless, existing GeoAI studies involving LLMs have primarily focused on entity recognition, semantic question answering or local relation extraction, with limited capacity to organize disaster events distributed across different times and locations into coherent evolution processes (Wang *et al.* 2024, Li *et al.* 2025). Moreover, without explicit geographic semantic constraints, LLMs may still generate semantically inconsistent or unreliable knowledge in complex disaster scenarios, thereby reducing the reliability of disaster knowledge modeling (Ji *et al.* 2025).

To strengthen the structured organization and semantic association of disaster knowledge, some studies have further integrated LLMs with KGs, using structured semantic constraints to improve knowledge organization and semantic consistency (Zhu *et al.* 2024). However, existing KG construction methods primarily adopt static entity–relation modeling and provide insufficient support for systematically representing the temporal sequence, spatial associations and dynamic evolution of disaster events (Zheng *et al.* 2022, Zhao *et al.* 2023). Many existing methods also rely on expert defined ontology structures and rule systems, which limits their adaptability when processing fragmented and continuously changing disaster information from social media (van Cauter and Yakovets 2024). Consequently, these methods still struggle to organize event information dispersed across different times and locations into coherent evolution sequences. Therefore, effectively integrating fragmented social media information and reconstructing the spatiotemporal evolution pathway of flood disasters while maintaining the semantic consistency and reliability of disaster knowledge remains an important research challenge.

3. Methodology

3.1. Knowledge-constrained LLMs for flood knowledge extraction

In this section, we present an LLM-based approach for extracting disaster-related entities and relationships. A FO integrated with geographic scene semantics is employed to constrain the extraction process and enhance the reliability of flood knowledge extraction.

3.1.1. Ontology modeling integrating geographic scene knowledge

To capture the spatiotemporal evolution of flood disasters and the coupling of multiple factors, this study constructs a Flood Scene Schema (FSS) centered on the Scene Unit. Based on geographic scene knowledge, the schema integrates six categories of semantic elements, including character, event, object, location, time and phenomenon,

Table 1. Semantic hierarchy of flood scene.

Level	Level description	Example
Entity layer	Define entity and event types in flood semantics based on six core semantic elements.	Entities: affected population, flood-inundated areas Events: dam breach, emergency response
Relation layer	*Spatial relations*: characterize spatial positions and topological relations	located in, adjacent to
	Temporal relations: describe event chronology and persistence	before, during, after
	Event association relations: describe semantic links among disaster events	$rain \overset{trigger}{\rightarrow} flood \overset{impact}{\rightarrow} damage$

into a unified structured space (Lü *et al.* 2022). It further characterizes complex flood scenes across six dimensions, including spatial location, geometric form, attribute properties, semantic description, inter-element relationships and evolutionary processes. The FSS jointly represents the spatial distribution, temporal progression and semantic associations among flood events, providing a spatially consistent and process interpretable knowledge framework. Its semantic structure is abstracted into two levels, namely entity and relation, as shown in Table 1.

Based on the FSS, a FO model is developed. The modeling process follows the principles specified in ISO/IEC 21838-1:2021, *Top-level ontologies (TLO) – Part 1: Requirements*, which provide general requirements for the design of top-level ontologies (International Organization for Standardization 2021). Classical ontology engineering methodologies are adopted to guide the modeling process (Grüninger and Fox 1995, Uschold and King 1995), and practical development guidelines are further applied to ensure interoperability, reusability and modularity (Noy and McGuinness 2001).

Specifically, ontology construction begins with a requirement analysis to define the semantic scope that the ontology model should cover. To ensure that the modeling objectives are well targeted and verifiable, a set of competency questions is formulated, which represents the key semantic questions that the ontology is expected to answer. On this basis, relevant concepts and relationships are derived from the attributes and relations defined in the FSS, and a structured ontology is modeled using the Web Ontology Language. The resulting flood ontology named FO structure can be formally defined as shown in the following equation:

$$\begin{cases} FO = \{C, R, A\} \\ R = R_s \cup R_t \cup R_e \\ A(c) = \{a_1, a_2, \ldots, a_n \mid a_i = (p_i, v_i)\} \end{cases} \tag{1}$$

where C denotes the set of concepts representing semantic entities in flood scenes, R denotes the set of semantic relations and A denotes the set of attributes. The relation set R includes spatial relations R_s, temporal relations R_t and event association relations R_e, which jointly describe the spatial, temporal and semantic connections among entities and events in flood scenes. In the attribute set A, a_i represents the ith attribute, p_i denotes the attribute name and v_i specifies the corresponding value domain. These attributes characterize semantic properties such as location, form, state and intensity.

3.1.2. Ontology-constrained knowledge graph construction driven by LLMs

This study introduces an ontology-constrained few-shot CoT strategy for LLM-driven KG construction, as shown in Figure 2. First, the extraction task is decomposed into ordered subtasks under the constraints of the FO. Accordingly, the original complex task is divided into subtasks with appropriate granularity and a clear logical order, including flood event identification, triggering relation extraction, and the extraction of affected objects and their attributes from social media data.

Next, few-shot prompting is employed with a small number of high-quality examples to help LLM recognize entity types, relations and structured output formats. CoT prompting is then used to guide step-by-step reasoning. The LLM first identifies and normalizes disaster-related entities, including events, locations and times, and generates node sets with unique identifiers. Based on these nodes, spatiotemporal and event relations are inferred to construct flood entity–relation triples, which are further linked back to the corresponding source tweets for evidence alignment. Candidate triples that lack explicit textual support are returned to the LLM for revision. This process

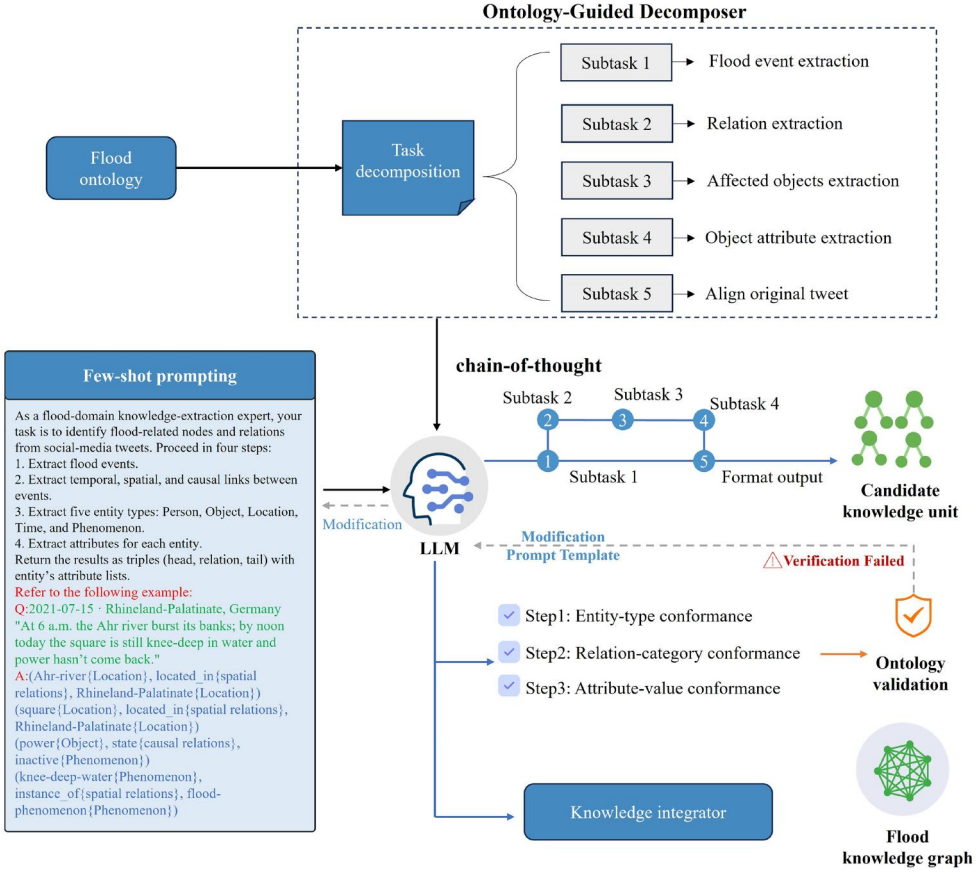


Figure 2. Ontology-constrained construction of a flood knowledge graph using few-shot CoT knowledge extraction into a sequence of ontology-guided subtasks. Few-shot CoT prompting enforces structured reasoning over entity types, relations and attributes. Candidate knowledge units are iteratively validated against FO constraints and corrected when violations occur, resulting in a semantically consistent and ontology-compliant flood knowledge graph.

ensures that evolutionary logic, temporal order and spatial associations are explicitly modeled during extraction, while keeping the extracted knowledge closely anchored to the original tweets and reducing hallucinated outputs.

Finally, candidate knowledge units are validated against the FO. In this process, entity types must conform to FO definitions, relation types are limited to spatial, temporal and event association categories, and attribute values, including time and location, are required to satisfy ontology constraints. If these constraints are violated, the corresponding candidates are returned to the LLM for correction until all subtasks pass ontology validation. After validation, the validated outputs are integrated to form a complete flood KG that supports querying and reasoning.

3.1.3. Cross-evaluation of LLMs for knowledge graph quality

The quality of flood KG extraction directly affects the reliability of subsequent evolution pathway reconstruction. Within the 'LLM-as-a-Judge' framework, this study designs an LLM cross-evaluation mechanism (Liu *et al.* 2023, Zheng *et al.* 2023). Evaluation is conducted across three core dimensions, namely accuracy, completeness and consistency. Each dimension is rated on a five-point Likert scale from L1 to L5. The evaluation workflow is illustrated in Figure 3. The scoring criteria are provided in Table 2.

Accuracy and completeness are the primary factors determining the reliability and coverage of KG extraction. Therefore, these two dimensions are each assigned a weight of 0.4. Consistency plays a supporting role in aligning extraction results across different LLMs or extraction batches and is assigned a weight of 0.2. Overall score S_{total} is calculated as a weighted sum, as defined in the following equation:

$$S_{\text{total}} = w_a \cdot S_{\text{acc}} + w_c \cdot S_{\text{com}} + w_s \cdot S_{\text{con}} \tag{2}$$

where S_{acc}, S_{com} and S_{con} denote the scores for accuracy, completeness and consistency, respectively. The corresponding weights w_a, w_c and w_s are set to 0.4, 0.4 and 0.2, respectively.

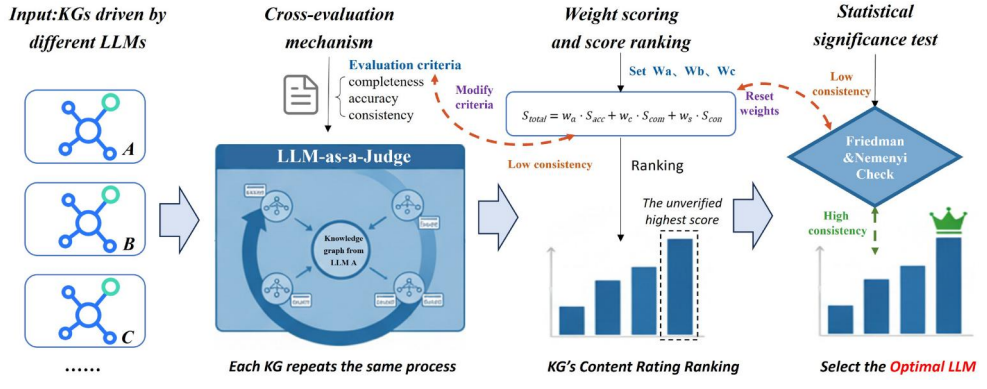


Figure 3. Cross-evaluation framework for assessing the quality of flood knowledge graphs generated by different LLMs. Knowledge graphs constructed by different LLMs are mutually evaluated within an LLM-as-a-Judge framework using three criteria: completeness, accuracy and consistency. Weighted aggregation yields overall quality scores and rankings. Friedman and Nemenyi tests are further applied to examine statistical significance and robustness, supporting the selection of the optimal LLM.

Table 2. Flood knowledge graph quality assessment criteria.

	Dimension		
Level	Completeness	Accuracy	Consistency
L1 (5 points)	All core entities, events and relations are correctly identified, with *complete coverage* and *no omissions*.	Extracted content *fully matches* the source text in semantic meaning and factual accuracy, with *no ambiguity*.	Results remain *consistent* across semantic representation, ontology constraints and LLM outputs.
L2 (4 points)	Most key knowledge is correctly identified, with only *minor secondary information missing*.	Extraction accuracy is *high*, with very few *minor deviations* or synonym-level errors.	Results are *largely consistent* in semantics and format, with *minor differences* that do not affect understanding.
L3 (3 points)	Most major knowledge units are extracted, but *some key entities or relations are missing*.	A *moderate level of semantic deviation* or misidentification is observed.	*Semantic differences* appear across LLMs or extraction runs, requiring *manual judgment*.
L4 (2 points)	Only limited information is identified, with *major content missing and insufficient coverage*.	*Misidentifications* and *factual inconsistencies* are frequent.	*Clear discrepancies* occur across LLM outputs, leading to *high semantic inconsistency*.
L5 (1 point)	Extraction results are fragmented and contain *little valid information or none*.	Most extracted content *does not align* with the source semantics or contains *serious errors*.	Results are *highly inconsistent* and lack a *coherent semantic framework*.

After scores are obtained for each dimension across all LLMs, the Friedman test is applied to assess overall differences among LLMs for both individual dimensions and the overall score. The Nemenyi post hoc test is then used to identify pairwise differences between LLMs. The corresponding formulations are provided in the following equations:

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1) \tag{3}$$

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6n}} \tag{4}$$

where χ_F^2 denotes the Friedman test statistic, n represents the number of evaluation instances, k is the number of compared LLMs, R_j denotes the sum of ranks for the jth LLM, CD denotes the critical difference, defined as the minimum average rank difference indicating a statistically significant difference in the Nemenyi post hoc test, q_α denotes the critical value for the Nemenyi test derived from the Studentized range distribution at significance level α for k compared LLMs.

A statistically significant difference between two LLMs exists if the difference in their average ranks exceeds the critical difference. If the test reveals substantial biases in how LLMs evaluate the same results, the prompt design or scoring rubrics should be refined accordingly. Conversely, if the observed differences primarily reflect the inherent performance of the LLMs, the evaluation results can be regarded as reliable for further analysis.

3.2. Reconstruction of main flood evolution pathway

3.2.1. Identification of main event chain in flood evolution

To identify the critical development pathway of flood disasters within the KG constructed in Section 3.1, multi-dimensional constraints are first integrated to generate candidate event chains. Subsequently, event importance is evaluated based on the structural characteristics of the FO. On this basis, the main event chain of flood evolution is further extracted from complex event associations, as shown in Figure 4.

First, candidate flood event chains are generated from the flood KG. Trigger-related phenomena or events are identified as starting nodes. From each starting node, a depth-first search (DFS) strategy is employed to traverse the graph, with the chains length incrementally expanded. Starting from node e_1, a candidate event chain is defined as $P = \langle e_1, e_2, \ldots, e_n \rangle$. During traversal, chain extension from event e_i to e_j is constrained by a tri-constraint set C, including semantic, temporal and spatial constraints. Under constraints, resulting evolution event chain can be formally expressed as follows:

$$\odot_1 \overset{C}{\Rightarrow} \odot_1 \rightarrow \odot_2 \rightarrow \odot_3 \rightarrow \odot_4 \rightarrow \cdots \rightarrow \odot_n \;\Leftrightarrow\; e_1 \xrightarrow{C=\{C_e, C_t, C_s\}} P = \langle e_1, e_2, \ldots, e_n \rangle$$

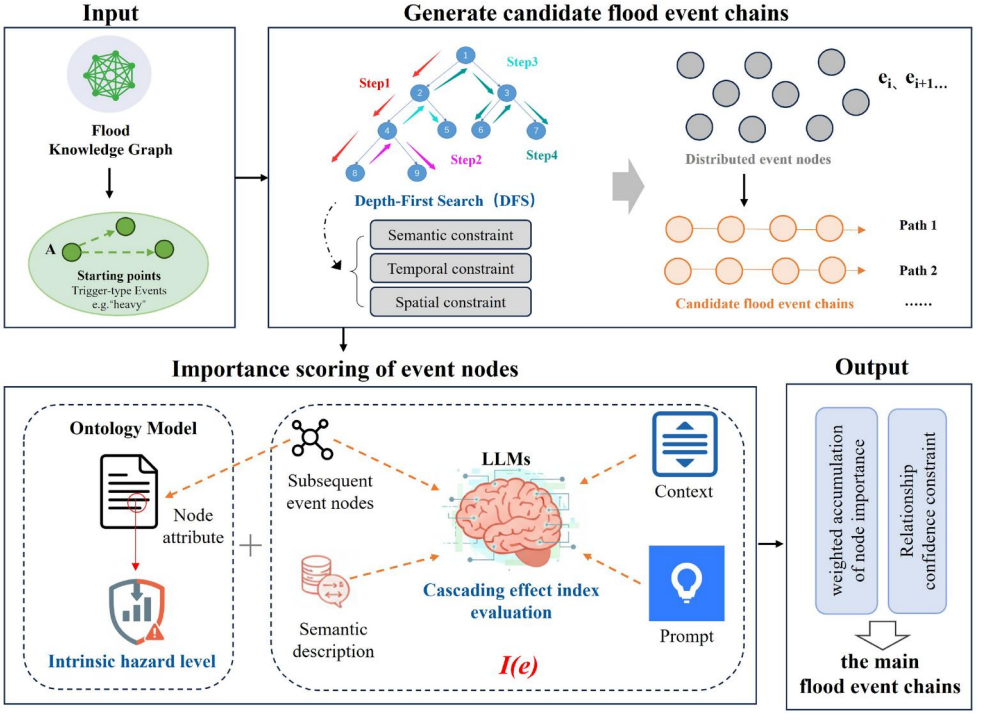


Figure 4. Identification of the main flood event chain from the flood knowledge graph. Trigger-type events are selected as starting points, and candidate chains are generated using depth-first search under event semantic and spatiotemporal constraints. Event importance is quantified by integrating ontology-based intrinsic hazard levels with cascading effects evaluated by LLMs. Candidate chains are ranked using a composite scoring function, yielding the dominant event chain that characterizes flood spatiotemporal evolution.

C_e represents the event semantic constraint, which restricts the semantic associations between candidate events. For any event pair (e_i, e_j), a candidate connection satisfies the semantic constraint only when an edge $r(e_i, e_j)$ exists in the KG and the relation belongs to the predefined types in the FO.

C_t represents the temporal constraint, which controls the chronological order and duration relationships between candidate events. For events with explicit timestamps, their occurrence times must satisfy $Time(e_i) \leq Time(e_j)$. For events with durations, temporal relations such as before, meet, overlap and contain are further used.

C_s represents the spatial constraint, which determines whether candidate events are spatially relevant in the context of flood disaster evolution. Formally, $IsRelevant(S(e_i), S(e_j))$ must be true, where $S(e_i)$ and $S(e_j)$ denote the spatial locations or spatial extents of events e_i and e_j, respectively.

This study evaluates spatial relevance from three aspects: place-hierarchy consistency, spatial-extent consistency and hydrological-unit consistency. The constraint definitions are summarized in Table 3.

Next, the importance of event nodes is evaluated. Traditional rule-based methods often struggle to capture implicit semantics and structural dependencies (Li *et al.* 2022a). By contrast, LLMs are used here to support semantic assessment and constraint-based reasoning across event attributes, spatial contexts and topological connections (Zhao *et al.* 2026). Accordingly, LLMs are employed to assess node importance and compute a comprehensive score, denoted as *I(e)*, as defined in the following equation:

$$I(e) = \alpha \cdot H_{\text{in}}(e) + \beta \cdot E_{\text{ca}}(e) \tag{5}$$

where $H_{\text{in}}(e)$ denotes intrinsic hazard level derived from ontology model. $E_{\text{ca}}(e) \in [0, 1]$ represents cascading effect index, which is a normalized score generated by the LLM

Table 3. Definition of constraints for evolution event chains expansion.

Constraint	Relationship	Formula	Description
C_e	Trigger	$r(e_i, e_j) \in R_e$	Event e_i triggers event e_j.
	Induce		Event e_i induces event e_j.
	Cause		Event e_i causes event e_j.
	Impact		Event e_i impacts event e_j.
C_t	Before	$Before(e_i, e_j)$	Event e_i occurs before event e_j.
	Meet	$Meet(e_i, e_j)$	Event e_i meets event e_j at a time point, while e_i occurs first.
	Overlap	$Overlap(e_i, e_j)$	Event e_i overlaps with event e_j for a certain period.
	Contain	$Contain(e_i, e_j)$	The time period of event e_i contains event e_j.
C_s	Place-hierarchy consistency	$IsRelevant(S(e_i), S(e_j))$	Events occur in the same place or in geographically nested places, such as a country–city or basin–valley relation.
	Spatial-extent consistency	$IsRelevant(S(e_i), S(e_j))$	The spatial extents of two events overlap, are adjacent, or belong to the same affected area.
	Hydrological-unit consistency	$IsRelevant(S(e_i), S(e_j))$	Events belong to the same flood-related hydrological unit, such as the same river valley, river reach, catchment, drainage unit or inundation zone.

based on semantic context and structural position of event e within the KG. Weights α and β reflect relative contributions of two components. Given chain-like propagation characteristics of flood disasters, these values are set to $\alpha = 0.3$ and $\beta = 0.7$.

The complete procedure is summarized in Algorithm 1.

Algorithm 1: Ontology-constrained evolution event chain reconstruction

Input: Flood knowledge graph $G = (E, R)$; Flood Ontology FO; maximum path length $L_{\max}$
Output: Candidate event chains set P; main evolution chain P^*

$P \leftarrow \emptyset$;
$T \leftarrow \text{ExtractTriggerEvents}(E, FO)$;
for each $e_{\text{start}} \in T$ **do**
  $p \leftarrow [e_{\text{start}}]$;
  DFS(p);
Function DFS(p):
  $e_i \leftarrow \text{LastNode}(p)$;
  $expanded \leftarrow false$;
  if $\text{Length}(p) \geq L_{\max}$ **then**
    $P \leftarrow P \cup \{p\}$;
    return;
  end
  for each $e_j \in E$ *such that* $e_j \notin p$ **do**
    $R_{ij} \leftarrow \text{GetRelations}(e_i, e_j, R)$;
    for each $r \in R_{ij}$ **do**
      if $r \in FO.R_{\text{event}} \wedge \text{Time}(e_i) \leq \text{Time}(e_j) \wedge \text{IsSpatiallyRelevant}(\text{Location}(e_i), \text{Location}(e_j))$ **then**
        $p' \leftarrow p \oplus e_j$; // appends e_j to path p;
        $expanded \leftarrow true$;
        DFS(p');
      end
  if $expanded = false \wedge \text{Length}(p) > 1$ **then**
    $P \leftarrow P \cup \{p\}$;
  end
for each $p \in P$ **do**
  Compute $S(p)$ using node importance and relation confidence;
$P^* \leftarrow \text{SelectTopPath}(P, S)$;
return P, P^*;

Finally, candidate chains are evaluated using a composite scoring function. At the chain level, each candidate chain receives a quantitative score to identify event chain that best represent the main evolutionary trajectory of flood disasters. The scoring function $S(P)$ incorporates two factors: weighted accumulation of node importance and confidence constraints of relations, as defined in Equation (6). Candidate chains are ranked according to $S(P)$, and the highest-scoring chain is selected as the main evolution event chain of flood disasters.

$$S(P) = \left(\sum_{i=1}^{n} \gamma^{i-1} I(e_i) \right) \cdot \left(\prod_{i=2}^{n} r_c(e_{i-1}, e_i) \right) \tag{6}$$

where γ denotes decay coefficient of node importance $(0 < \gamma \leq 1)$, which controls attenuation of node contributions as chain depth increases. When γ is small, importance of upstream triggering events is amplified, whereas contributions of secondary events in the middle and later stages decrease rapidly with increasing depth. This formulation emphasizes dominant role of driving and triggering events in flood disaster event chain. The term $r_c(e_{i-1}, e_i)$ represents the confidence of the event association between two adjacent event nodes, which is jointly determined by temporal consistency, spatial consistency and event semantic consistency.

3.2.2. Spatiotemporal completion of flood evolution pathway

The main evolution pathway inferred using the method described in Section 3.2.1 reveals the underlying logic and key structural patterns of flood disaster evolution. However, this pathway remains an abstract representation composed of symbolic nodes and semantic relationships, and its correspondence with real-world geographic spatiotemporal processes has not been explicitly established. To bridge this gap and endow the inferred results with locatable and evolvable physical characteristics, this section enriches event nodes with explicit spatiotemporal attributes. Through this enrichment, the main evolution pathway is extended from a purely semantic representation to the geographic semantic level.

Each event node e_i in the main event chain is examined, and its pre-extracted spatiotemporal attributes are retrieved from the flood KG. Temporal attributes include the event timestamp, *start_time* and *end_time*, while spatial attributes comprise the country and city of occurrence as well as the corresponding central geographic coordinates.

These spatiotemporal attributes are integrated with event types, textual descriptions and event association. The integration yields a chronologically ordered sequence of structured data objects, which are defined as spatiotemporal evolution pathway, as formalized in the following equation:

$$P_{\mathrm{st}} = \langle e_1(t_1, s_1) \xrightarrow{R_e} e_2(t_2, s_2) \xrightarrow{R_e} \cdots \xrightarrow{R_e} e_n(t_n, s_n) \rangle \tag{7}$$

where e_i represents the ith event node, t_i and s_i denote its temporal and spatial attributes, respectively. R_e denotes the semantic association between adjacent event nodes. The resulting pathway explicitly describes when and where events occur, as well as what happens during the disaster evolution process. In this way, semantic information is integrated with spatiotemporal information within a unified representation.

3.3. Augmented representation of flood disaster evolution

Narrative representation organizes complex dynamic processes through temporal order, event associations and semantic coherence. To intuitively present the reconstructed main flood evolution pathway in 3D scenes, this study adopts narrative progression as the organizing logic and develops an augmented representation method integrating event symbols, temporal control and viewpoint scheduling (Li *et al.* 2021, 2024).

First, a 3D symbol system is designed based on the syntax–semantics–pragmatics framework of semiotics (Morris 1938) and map-based storytelling (Roth 2021). Flood disaster evolution is represented as a sequence of discrete events using the following equation:

$$\begin{cases} e_i = \{t_i, s_i, a_i\}, \\ s_i = (x_i, y_i, z_i), \\ a_i = \{a_{i1}, a_{i2}, \ldots\} \end{cases} \tag{8}$$

where t_i denotes the time at which an event occurs, s_i represents spatial location composed of spatial coordinates and a_i denotes event attribute vector.

The symbol system comprises three levels. At the syntactic level, visual variables, including shape, color, size, transparency and motion, distinguish different flood events. At the semantic level, symbols are mapped to corresponding entities and encode specific flood information. At the pragmatic level, textual descriptions, images, chart summaries, flashing effects and annotations guide users to recognize event associations and evolution trends.

Based on this symbol system, the reconstructed spatiotemporal evolution pathway is used as the primary narrative thread for organizing flood events. Time t serves as the driving variable, and a control function $S(t)$ jointly regulates event visibility and observation viewpoints (Li *et al.* 2022b), as shown in the following equation:

$$\mathcal{S}(t) = \begin{cases} Et_0, P(x,y,z), R(h,p,r), t = t_0, \\ Et_1, P(x,y,z), R(h+s,p,r), t = t_1. \end{cases} \tag{9}$$

where E_{t0} and E_{t1} denote the invisible and visible states of an event before and after its occurrence, respectively. $P(x,y,z)$ defines the viewpoint position, while $R(h,p,r)$ specifies its yaw, pitch and roll. The parameter s controls the rotational increment of the viewpoint over time.

By coordinating event visibility and viewpoint transitions, the proposed method provides a continuous and controllable 3D representation of flood evolution pathway, as shown in Figure 5. It supports the structured presentation of flood events and facilitates the identification of key stages, spatial expansion patterns and event associations during disaster evolution.

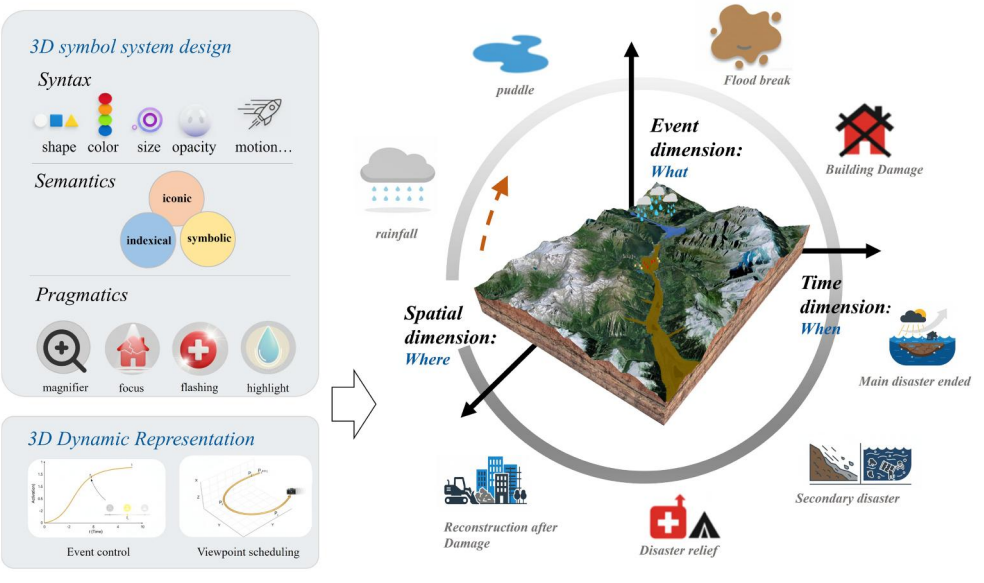


Figure 5. Knowledge-guided 3D representation of flood spatiotemporal evolution. Flood events inferred from KG are mapped into a 3D scene through a symbol system spanning syntax, semantics and pragmatics. Events are organized along spatial, temporal and event dimensions. Dynamic event control and viewpoint scheduling ensure temporal coherence and narrative continuity, enabling an interpretable representation of flood disaster evolution.

4. Experiments and results

4.1. Case study and datasets

This study adopts the extreme European flood event of July 2021 as the case study. Semantic information on disaster evolution was obtained from flood tweets collected using Twint between 1 July 2021 and 31 July 2021. A total of 34,731 tweets were initially collected. After standardized preprocessing, including noise removal and NLTK stop-word filtering, 20,962 high quality texts were retained for knowledge extraction, as shown in Figure 6. To support 3D scene construction and process simulation, multi-source geographic environmental data were also integrated. These data include DEM data with one meter resolution for flood simulation and terrain modeling, administrative boundary data for defining spatial extents, and ERA5 precipitation data for meteorological background analysis. Detailed descriptions of the datasets are provided in Table 4.

To further understand the composition of the cleaned tweet corpus, we conduct a content distribution analysis. At the ontology-element level, tweets containing disaster events or phenomena (event, phenomenon) account for 99.34%, tweets with explicit temporal or spatial references (time, location) are 98.75%, and 41.07% tweets involve specific affected objects or related people(object, character), indicating that social



(a) Frequent disaster events and phenomena (b) Frequent temporal and spatial references (c) Frequent affected objects and populations

Figure 6. Word clouds of high-frequency expressions in the cleaned flood-tweet corpus. Panels (a)–(c) show frequent disaster events and phenomena, temporal and spatial references, and affected objects and populations, respectively. Word size is proportional to term frequency, illustrating the dominant semantic information contained in the tweets.

Table 4. Overview of research datasets.

Data type	Spatial and temporal coverage	Data source	Primary use
Tweet data	Europe, July 2021	Twitter	Extraction of disaster-related semantic information
DEM data	Ahr Valley, Rhineland–Palatinate, Germany, at 1 m resolution	GeoShop Rheinland-Pfalz[a]	Flood simulation and high-resolution 3D scene construction
Administrative boundaries	European country level	Eurostat GISCO[b]	Study area delineation and spatial referencing
Flood simulation results	Areas associated with disaster chains	Cellular automata model developed in this study	Characterization of flood spatiotemporal evolution
Precipitation data	Europe	Copernicus Climate Data Store ERA5 dataset[c]	Meteorological background analysis

[a]https://geoshop.rlp.de/.
[b]https://ec.europa.eu/eurostat/web/gisco/geodata.
[c]https://cds.climate.copernicus.eu.

media data provide dense event, spatiotemporal and affected object information that can support the identification of disaster spatiotemporal evolution.

4.2. Prototype system implementation

This study developed a 3D flood spatiotemporal representation system deployed on a Windows 11 Professional platform. The experimental setup includes an AMD Ryzen 7 5800X processor operating at 3.8 GHz, an NVIDIA GeForce RTX 3070 graphics card and 32 GB of DDR4 memory. The system adopts a modular architecture composed of a KG module, an evolution pathway reconstruction module and a disaster evolution representation module. It supports front-end representation of the KG together with real-time evolution pathway reconstruction. Flood disaster evolution is presented from a spatiotemporal perspective. The prototype system interface is shown in Figure 7.

4.3. Experimental design

To systematically evaluate the reliability of the proposed method for disaster knowledge extraction and evolution pathway reconstruction, as well as the interpretability of the 3D flood representation, this study randomly sampled 1100 tweets from the cleaned twitter dataset for three progressive evaluation experiments.

4.3.1. Knowledge extraction performance evaluation and comparative experiments

This experiment evaluates the knowledge extraction method proposed in Section 3.1 and examines its applicability in real disaster contexts. Four representative LLMs from different model families, namely DeepSeek-V3.2, GPT-4o-Mini, Gemini-2.5-Flash and Qwen3.5-Flash, are selected for comparison. All models adopt the same FO-constrained

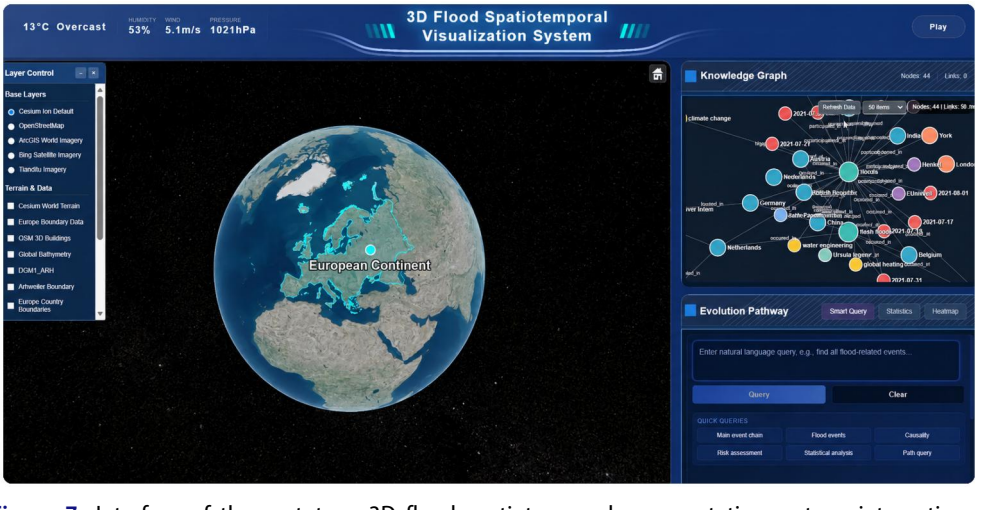


Figure 7. Interface of the prototype 3D flood spatiotemporal representation system, integrating a globe scene viewer with a knowledge graph panel and an evolution pathway reconstruction workspace for interactive analysis.

few-shot CoT prompting strategy to ensure a fair comparison of disaster semantic understanding and structured knowledge generation.

Comparative experiments are then conducted using the LLM with the best overall performance. Three prompting configurations are compared to analyze the effects of explicit reasoning and domain ontology constraints on knowledge extraction performance.

i. Zero-shot prompting without examples, reasoning chains or ontology constraints, serving as the baseline.
ii. Few-shot CoT prompting, which introduces expert-annotated examples from the separate development set. Specifically, 30 tweets with varying levels of information content (10 low, 10 medium and 10 high) are selected, along with their corresponding extraction processes and results, to guide LLM-based knowledge extraction without applying domain ontology constraints.
iii. FO-constrained few-shot CoT prompting, which uses the same few-shot examples as (ii) and further embeds the conceptual system and relation constraints defined by the FO.

4.3.2. Expert-based validation of evolution pathway reconstruction

To assess the rationality and reliability of the disaster evolution pathway reconstruction method, an expert consensus-based evaluation experiment is conducted. Based on the constructed flood KG, an evaluation set containing 120 candidate evolution pathways is generated. Among them, 60 pathways are produced by the proposed constrained reconstruction mechanism and constitute the Constrained Group, while the remaining 60 pathways are obtained through unconstrained search and form the Unconstrained Baseline Group.

In addition to the three experts involved in knowledge annotation, two independent domain experts participate in the evaluation. All experts rate the evolution chains from both groups using a five-level Likert scale across three dimensions, including evolution logic soundness, completeness and spatiotemporal consistency. To avoid evaluation bias, all candidate pathways are anonymized and randomly shuffled before evaluation, so that experts do not know the group source of each pathway. Statistical comparison of the ratings is used to examine whether the proposed method more effectively identifies key disaster evolution pathway.

4.3.3. User evaluation of 3D representation

To assess the effectiveness of augmented 3D flood representation for disaster cognition and understanding, a questionnaire-based user study is conducted. The main flood disaster evolution pathway inferred in Section 3.2 serves as input for generating three different forms of disaster evolution representations. These representations are compared to evaluate their impact on user understanding of flood disaster evolution.

The experiment involves 90 participants aged between 20 and 50, with educational backgrounds ranging from high school to doctoral level. All participants come from non-specialist backgrounds and have not received systematic training in GIS, flood-related domains or 3D representation. A controlled experimental design is adopted,

and participants are randomly assigned to three groups, each containing 30 individuals.

i. Text-based Group, which provides only KG-based and text-based descriptions of flood disaster evolution chains.
ii. 3D Process Group, which presents 3D dynamic flood disaster evolution without explicit event chain annotation.
iii. Full Representation Group, which applies the complete 3D flood disaster chain representation proposed in this study. Key event nodes, directional event associations and temporal relationships are explicitly annotated within the dynamic scene.

After completing the assigned tasks, participants rate Likert-scale questionnaire items related to disaster evolution understanding. The collected responses are analyzed to evaluate differences in perceived disaster cognition across the three information presentation modes.

4.4. Results and analysis

4.4.1. Evaluation of knowledge extraction performance

Under a unified FO constraint and few-shot CoT prompting framework, four mainstream LLMs are first evaluated for flood-related knowledge triple extraction from social media tweets. At the subjective evaluation level, the cross-evaluation mechanism described in Section 3.1.3 is used to assess output quality across Accuracy, Completeness, Consistency and Aggregate Score. To avoid potential self-preference, self-evaluation results generated by the LLMs are excluded. As shown in Figure 8, scoring trends across evaluating LLMs are largely consistent, indicating that the cross-evaluation mechanism is stable at the subjective level. Gemini-2.5-Flash obtains the highest overall score and leads in consistency and completeness, while DeepSeek-V3.2 performs best in accuracy but shows lower consistency, which limits its overall knowledge extraction performance.

Figure 9 further validates differences among LLMs in Aggregate Score, Accuracy, Completeness and Consistency from a statistical perspective. Friedman test results indicate significant differences among models across all dimensions ($p < 0.001$). Nemenyi post hoc comparisons show that Gemini-2.5-Flash significantly outperforms GPT-4o-Mini, Qwen3.5-Flash and DeepSeek-V3.2 in Aggregate Score ($p < 0.01$).

At the objective evaluation level, a manually annotated benchmark dataset containing 1100 randomly sampled tweets is constructed. Three domain experts independently annotate flood-related triples according to a unified annotation protocol and ontology constraints. The inter-annotator agreement measured by Fleiss' Kappa is 0.76, indicating substantial agreement (Landis and Koch 1977). For samples with disagreement, majority voting and expert discussion are used to form the final gold-standard dataset. Based on this dataset, Precision, Recall, *F*1-score and Hallucination Rate (HR) are calculated for each LLM, as shown in Figure 10. HR measures the proportion of triples generated by the model but not supported by the source tweet

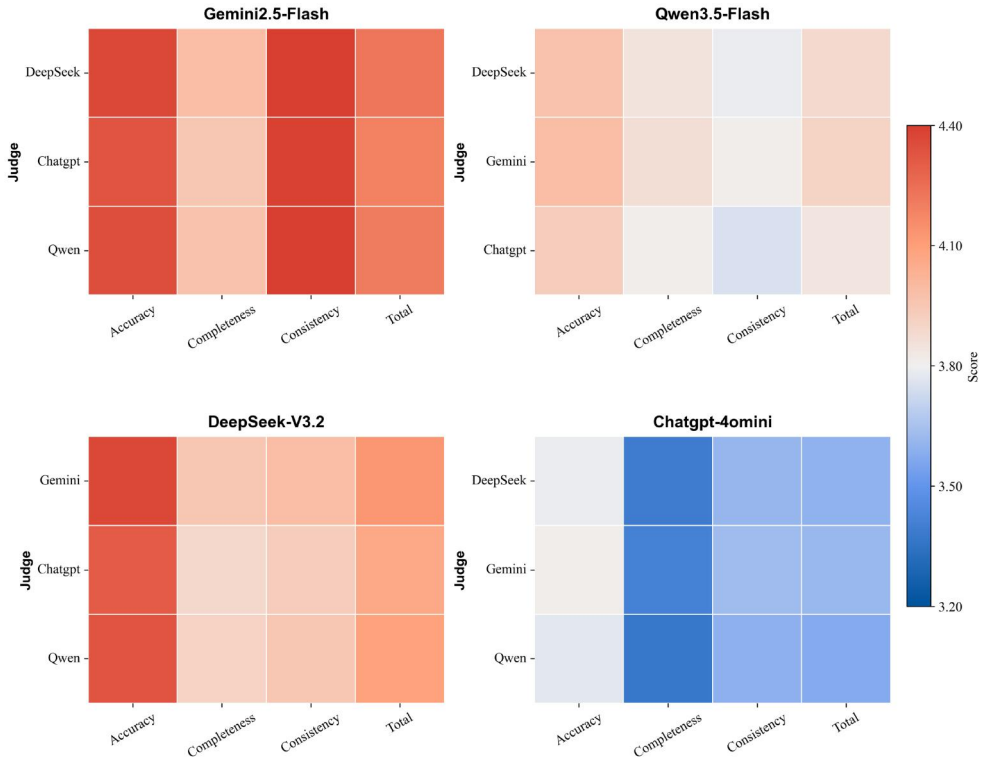


Figure 8. Cross-evaluation heatmaps of the quality of flood knowledge graphs generated by different LLMs under the LLM-as-a-Judge framework. The four panels present the evaluation results for the knowledge graphs generated by Gemini-2.5-Flash, Qwen3.5-Flash, DeepSeek-V3.2 and GPT-4o-Mini, respectively, in terms of accuracy, completeness, consistency and aggregate score. Warmer colors indicate higher evaluation scores.

evidence. Gemini-2.5-Flash achieves the best Precision, Recall and *F*1-score, while also obtaining the lowest HR (0.09), indicating strong overall knowledge extraction performance. This result is consistent with the subjective evaluation trend.

Based primarily on the manually annotated benchmark results and objective metrics, and with the LLM cross-evaluation used as supplementary semantic evidence, Gemini-2.5-Flash is selected for comparative experiments. Zero-shot, few-shot CoT and few-shot CoT combined with FO constraints are compared to analyze the incremental contribution of each prompting strategy. The overall trend is shown in Figure 11, and detailed comparison results are reported in Table 5.

Results in Figure 11 and Table 5 show that, compared with zero-shot prompting, few-shot CoT clearly improves Recall (+0.16) and *F*1-score (+0.10), indicating that few-shot chain reasoning helps identify flood disaster semantic information more comprehensively. After FO constraints are further embedded, Precision (+0.12) and Recall (+0.10) continue to increase, the *F*1-score reaches 0.83, and HR decreases substantially to 0.09. This suggests that ontology constraints can effectively suppress hallucinated generation and enhance knowledge consistency. Bootstrap resampling with 1000 iterations further estimates a 95% confidence interval of [0.81, 0.84] for the *F*1-score of the

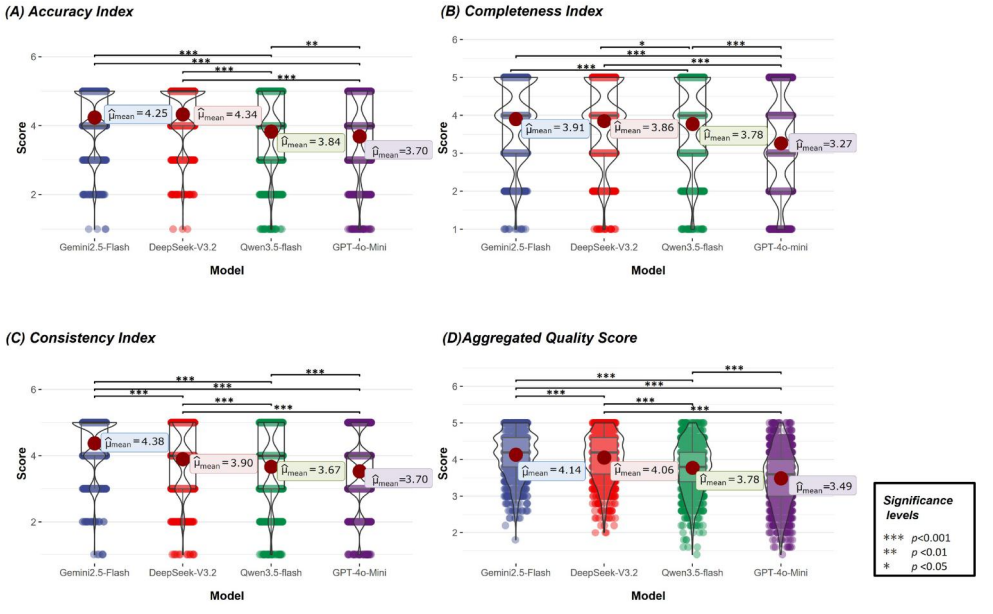


Figure 9. Comparison of LLM performance under multiple evaluation dimensions. Panels illustrate the distributions of different LLMs across aggregated quality, completeness, accuracy and consistency metrics.

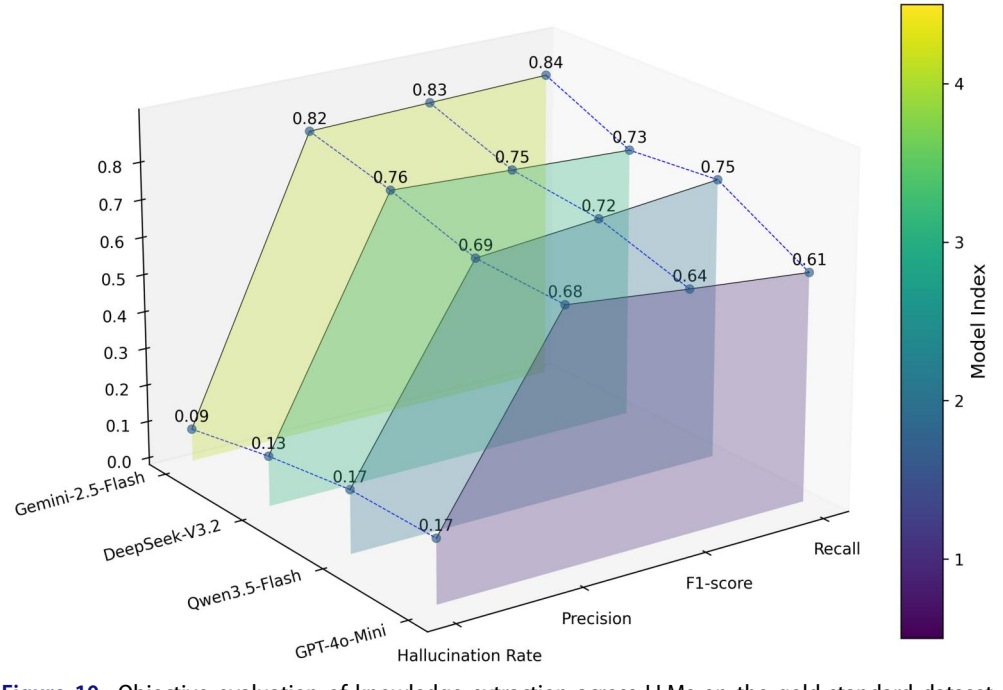


Figure 10. Objective evaluation of knowledge extraction across LLMs on the gold-standard dataset, comparing Precision, Recall, *F*1-score and Hallucination Rate.

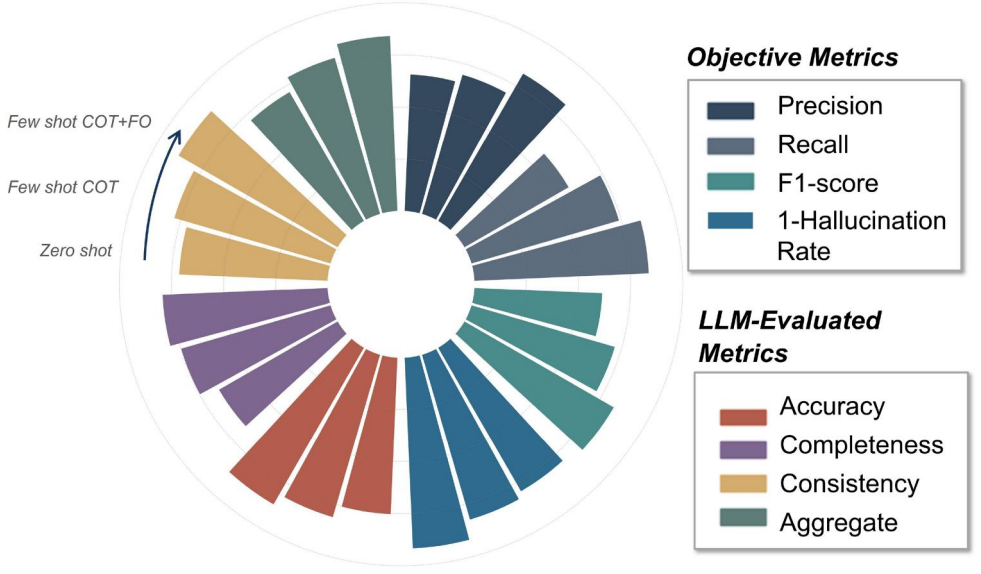


Figure 11. Multi-dimensional comparison of prompting strategies for flood knowledge extraction across objective and LLM-evaluated metrics. The figure compares three prompting strategies, namely zero-shot, few-shot CoT and few-shot CoT with FO constraints. Results are arranged clockwise from zero-shot to few-shot CoT and few-shot CoT + FO, reflecting increasing levels of reasoning complexity and ontology constraint.

Table 5. Comparative analysis of prompting strategies for knowledge graph construction.

Prompt strategy	Precision	Recall	*F*1-score	Hallucination Rate
Zero-shot	0.66	0.58	0.62	0.20
Few-shot CoT	0.70 (+0.04)	0.74 (+0.16)	0.72 (+0.10)	0.17 (−0.03)
Few-shot CoT + FO	**0.82 (+0.12)**	**0.84 (+0.10)**	0.83 (+0.11)	**0.09 (−0.08)**

Prompt strategy	Accuracy	Completeness	Consistency	Aggregate
Zero-shot	3.72	3.25	3.54	3.50
Few-shot CoT	4.01 (+0.29)	3.68 (+0.43)	3.87 (+0.33)	3.85 (+0.35)
Few-shot CoT + FO	**4.25 (+0.24)**	**3.91 (+0.23)**	**4.38 (+0.51)**	**4.14 (+0.29)**

Note: Bold values indicate the best performance among the three prompting strategies for each metric.

Gemini few-shot CoT + FO strategy, indicating stable extraction performance under ontology constraints.

The subjective evaluation results show a consistent pattern. Few-shot CoT + FO obtains the highest scores in Accuracy, Completeness and Consistency, with an Aggregate Score of 4.14, clearly outperforming the other two strategies. Overall, few-shot CoT improves knowledge reasoning, while ontology constraints define semantic boundaries, suppress hallucinations and enhance consistency. Their joint use enables more accurate, complete and consistent extraction of flood disaster knowledge.

Based on the optimal strategy, full-scale knowledge extraction and deduplication are performed on all flood tweets, resulting in a flood KG containing 9171 triples and 6393 nodes across six core semantic categories (Table 6). This KG provides a data foundation for subsequent evolution pathway reconstruction and 3D scene representation.

Table 6. Semantic entity composition of the flood knowledge graph.

Type	Number
Event	2600
Location	1739
Object	1371
Character	428
Phenomenon	199
Time	56

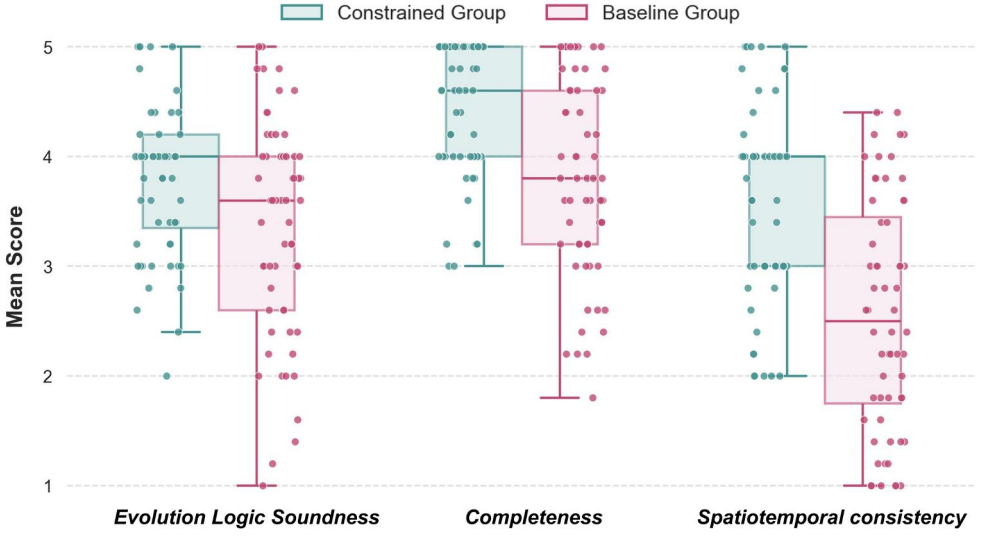


Figure 12. Expert evaluation of evolution pathway reconstruction results. Boxplots compare the Constrained Group and the Unconstrained Baseline Group across evolution logic soundness, completeness and spatiotemporal consistency. Each overlaid scatter point represents the mean score of one candidate pathway averaged across the five experts.

Table 7. ICC results of expert ratings.

Dimension	ICC(2,1)	95% CI	*p* Value	Reliability
Evolution logic soundness	0.808	[0.73, 0.87]	$p < 0.001$	Good
Completeness	0.798	[0.71, 0.87]	$p < 0.001$	Good
Spatiotemporal consistency	0.903	[0.86, 0.94]	$p < 0.001$	Excellent

4.4.2. Results and analysis of flood evolution pathway reconstruction

Five domain experts independently evaluate the evolution chain reconstruction results of the Constrained Group and the Unconstrained Baseline Group, as shown in Figure 12. Overall, the Constrained Group outperforms the baseline group across evolution logic soundness, completeness and spatiotemporal consistency. The distribution of expert ratings also shows that the proposed method yields more concentrated high-score results, indicating substantially reduced reconstruction uncertainty and improved stability and consistency.

To further examine the consistency of expert ratings, the intraclass correlation coefficient (ICC) is used to assess inter-expert agreement for the Constrained Group. As reported in Table 7, expert ratings show high consistency across all three evaluation dimensions.

Table 8. Attributes of the main event nodes in the reconstructed flood disaster chain.

Event node	Main attributes
Persistent heavy rainfall	*Time*: 14–15 July 2021; *Region*: Western Europe; *Phenomenon*: persistent rain, heavy rainfall, severe weather
River level rise	*Location:* Ahr River and nearby affected valleys; *Phenomenon:* rising floodwaters, rapidly increasing water level; *Spatial cue:* river corridor and valley areas
River overflow and flash flooding	*Location:* Ahr Valley and riverside settlements; *Event type:* river overflow, flash flood, fast-moving floodwater; *Spatial pattern:* floodwater spreading from river channels into nearby settlements
Urban inundation	*Affected areas:* riverside towns, low-lying built-up areas, valley-floor settlements; *Affected objects:* streets, roads, houses, vehicles, bridges, public facilities
Infrastructure and public facilities	*Damaged objects:* roads, bridges, railways, residential buildings, transport facilities, power supply, communication services
Residents and communities	*Affected people:* trapped residents, missing persons, casualties, displaced households; *Community impact:* evacuation needs, shelter demand, disruption of daily life
Emergency response and evacuation	*Actors:* police, firefighters, rescue teams, local authorities, residents; *Actions:* evacuation warnings, rescue operations, search and rescue, temporary shelter support
Post-flood recovery	*Stage:* after floodwater recession; *Actions:* cleanup, mud removal, house repair, infrastructure restoration, community recovery

The reconstructed main flood disaster evolution pathway shows a clear event-level evolutionary structure and can be summarized as follows, with the corresponding attributes reported in Table 8.

$$\odot_1 \text{ Persistent Heavy Rainfall} \xrightarrow{\text{triggered}} \odot_2 \text{ River Level Rise} \xrightarrow{\text{led to}}$$

$$\odot_3 \text{ River Overflow and Flash Flooding} \xrightarrow{\text{caused}} \odot_4 \text{ Urban Inundation} \xrightarrow{\text{damaged}}$$

$$\odot_5 \text{ Infrastructure and Public Facilities} \xrightarrow{\text{affected}} \odot_6 \text{ Residents and Communities} \xrightarrow{\text{prompted}}$$

$$\odot_7 \text{ Emergency Response and Evacuation} \xrightarrow{\text{transitioned to}} \odot_8 \text{ Post-flood Recovery}$$

To ensure the traceability of the reconstruction results, each event node was further linked back to the original tweet records for evidence alignment. After completing evolution pathway reconstruction based on the flood KG, post-event investigation reports and related event studies (Kreienkamp *et al.* 2021, Tradowsky *et al.* 2023) were further used for supplementary comparison. This comparison examined whether the reconstructed pathway was consistent with post-event synthesized understanding in terms of key stages and evolutionary logic. It serves as an external plausibility reference for interpreting the reasoning results. The results indicate that the proposed framework can organize fragmented real-time disaster information into a coherent flood disaster event structure under temporal, spatial and semantic constraints.

4.4.3. 3D representation and user evaluation results

This study integrates flood KG and evolution pathway reconstruction results, together with high-resolution terrain data and flood simulation outputs, to represent flood disaster evolution in a 3D spatiotemporal space. The representation depicts key event-level stages, including persistent heavy rainfall, river level rise, river overflow and flash flooding, urban inundation, infrastructure and public facility disruption, affected residents and communities, emergency response, and post-flood recovery, through

formalized symbols and dynamic inundation processes. Core event nodes are connected according to temporal order and event evolutionary direction, as shown in Figure 13. To improve accessibility and facilitate understanding of the full flood disaster evolution, the 3D representation is recorded as a video and shared via YouTube[2] and Bilibili.[3]

Following the user experiment design in Section 4.3.3, this study presents flood disaster evolution representations to participants using three information presentation modes, with examples shown in Figure 14. All three modes use identical data sources, flood simulation results and temporal coverage. They differ only in information organization and in how event semantics are conveyed. To ensure consistency in user perception and comparison, the presentation duration of all three representation modes was standardized to 2 min 40 s, corresponding to the duration of the enhanced 3D representation.

The questionnaire evaluates overall understanding of flood disaster evolution, clarity of evolutionary relationships, and comprehension of temporal consistency, as summarized in Table 9.

As shown in Table 10, Cronbach's α values for Q1, Q2 and Q3 are 0.693, 0.746 and 0.691, respectively, indicating acceptable to good internal consistency. The Kaiser–Meyer–Olkin (KMO) value reaches 0.902, indicating excellent sampling adequacy; Bartlett's test of sphericity is significant ($p < 0.001$), showing strong correlations among variables and supporting factor analysis. Overall, the questionnaire data show good reliability and structural suitability for subsequent analysis.

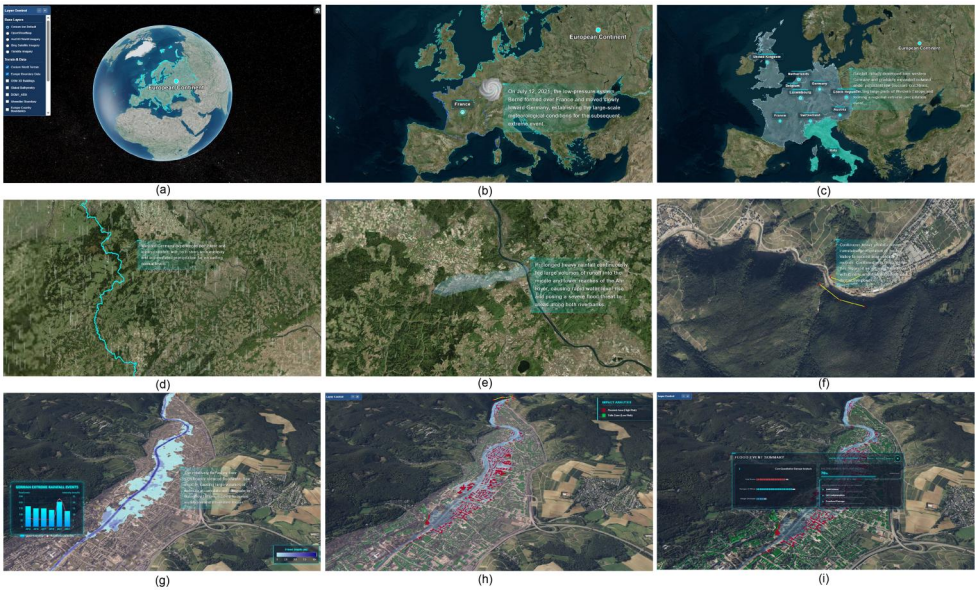
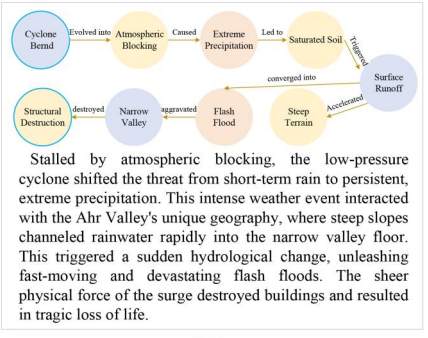


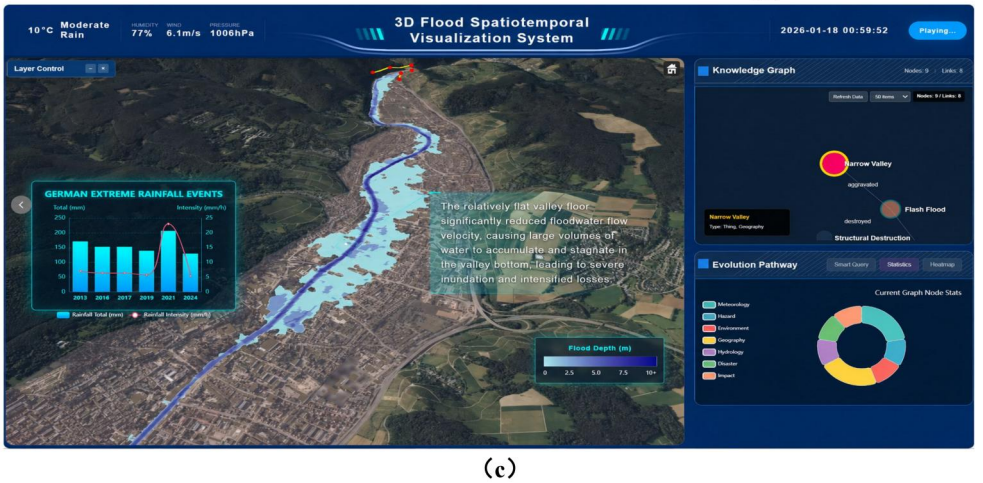
Figure 13. 3D spatiotemporal representation of the inferred flood disaster evolution. Panels (a)–(i) illustrate the progressive representation of the inferred event-level flood disaster chain within a 3D geographic scene, ranging from persistent heavy rainfall and hydrological response to urban inundation, infrastructure disruption, affected communities, emergency response and post-flood recovery.



(a)



(b)



(c)

Figure 14. Representation examples used in the user study under three presentation modes. Panel (a) corresponds to Group A, which provides knowledge-graph and text-based evolutionary descriptions. Panel (b) corresponds to Group B, which presents 3D flood disaster evolution representation without explicit annotation. Panel (c) corresponds to Group C, which applies the full evolutionary knowledge-enhanced 3D representation integrating event-chain cues and analytical panels.

Table 9. Questionnaire for evaluating flood disaster representation.

Dimension	Question description	Scale
Q1: Overall understanding	U1: I can understand the overall evolution of the flood disaster from onset to development	1: Not at all 2: Slightly 3: Moderately 4: Fairly 5: Very.
	U2: The representation facilitates identification of key stages in flood disaster evolution	
	U3: I have formed a comprehensive understanding of flood disaster evolution	
Q2: Evolutionary clarity	C1: I can clearly identify the primary triggering factors of the flood disaster	
	C2: I can distinguish between causal events and resulting events during disaster evolution	
	C3: The representation facilitates understanding of evolutionary relationships among disaster events	
Q3: Temporal consistency	T1: I can correctly determine the temporal order of disaster-related events	
	T2: The representation supports understanding of the temporal progression of disaster evolution	
	T3: My understanding of event chronology is consistent with the representation	

Table 10. Reliability and factorability assessment of the questionnaire.

Category	Scale/test	Key statistics	Interpretation
Reliability	Q1	Cronbach's $\alpha = 0.693$	Acceptable internal consistency
	Q2	Cronbach's $\alpha = 0.746$	Good internal consistency
	Q3	Cronbach's $\alpha = 0.691$	Acceptable internal consistency
	Overall scale	Cronbach's $\alpha = 0.876$	Excellent overall reliability
Validity	KMO measure	KMO = 0.902	Sampling adequacy is meritorious
	Bartlett's test	$\chi^2 = 291.88$, $df = 36$, $p < 0.001$	Correlation matrix suitable for factor analysis

Table 11. Mean scores of three participant groups across questionnaire dimensions.

Group	Q1, mean ± SD	Q2, mean ± SD	Q3, mean ± SD	Overall, mean ± SD
A	2.80 ± 0.71	2.98 ± 0.75	2.78 ± 0.74	2.85 ± 0.64
B	3.20 ± 0.60	3.56 ± 0.58	3.41 ± 0.54	3.39 ± 0.45
C	3.79 ± 0.58	4.03 ± 0.54	3.83 ± 0.65	3.89 ± 0.51
Kruskal–Wallis *H*	28.14	29.65	26.75	35.94
p Value	< 0.001	< 0.001	< 0.001	< 0.001
η^2	0.30	0.32	0.28	0.39

As shown in Table 11, different information presentation modes exhibit significant differences across all evaluation indicators, including the three cognitive dimensions and the overall evaluation (Kruskal–Wallis test, $p < 0.001$). The Bonferroni-adjusted Dunn's post hoc tests further indicate that the scores of Group C are significantly higher than those of Group A in Q1, Q2, Q3 and the overall evaluation (all adjusted $p < 0.001$). In addition, Group C achieves significantly higher scores than Group B in Q1 ($p = 0.008$), Q2 ($p = 0.012$) and the overall evaluation ($p = 0.007$). Group B also scores significantly higher than Group A in Q2 ($p = 0.032$), Q3 ($p = 0.008$) and the overall evaluation ($p = 0.010$), whereas the difference between Groups A and B in Q1 is not significant ($p = 0.066$).

When only KGs and textual descriptions are provided (Group A), participants obtain relatively low scores in overall understanding, evolutionary clarity and spatiotemporal consistency, indicating the limited cognitive support provided by this information presentation mode. After introducing 3D representation without symbolic annotation (Group B), the mean scores across the three cognitive dimensions increase, suggesting that spatialized representation can enhance users' understanding of disaster evolution to some extent. In comparison, the proposed knowledge-guided enhanced representation method (Group C) achieves the highest mean scores across all dimensions, with a particularly pronounced improvement in evolutionary clarity. These results suggest that the proposed method is especially effective in supporting users' understanding of event relationships in disaster evolution, while also improving the overall cognitive evaluation.

Figure 15 presents the score distributions across cognitive dimensions for the three information presentation modes. Clear separation is observed among the methods, indicating systematic differences in users' cognitive responses. Across all dimensions, the proportion of higher ratings increases progressively from Group A to Group C, while the proportion of lower ratings decreases accordingly. This trend suggests that the increasing integration of three-dimensional representation and event semantics enhances users' understanding of flood disaster evolution. Overall, the proposed 3D augmented flood representation method strengthens the intuitive representation of

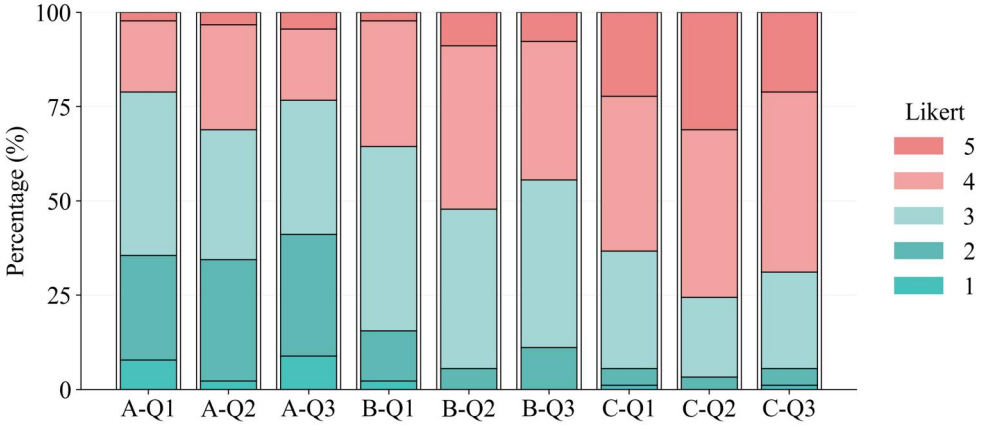


Figure 15. Distribution of user ratings across cognitive dimensions for the three participant groups (A–C). Stacked bars show the proportions of Likert-scale responses for overall understanding (Q1), evolutionary clarity (Q2) and temporal consistency (Q3).

the full flood disaster evolution and more effectively conveys event relationships, resulting in a marked improvement in users' overall cognitive understanding of flood disasters.

5. Discussion

This study proposes an augmented representation method for the spatiotemporal evolution of flood disasters based on large-scale fragmented social media data. It enables the automatic construction of a flood KG, reconstruction of the main disaster evolution pathway, and 3D representation of flood scenes, thereby providing a new technical approach for public understanding and risk communication of complex disaster processes. However, several limitations should also be noted.

First, social media data are essentially social sensing information. They often reflect local disaster conditions, public perception or news dissemination rather than complete disaster evolution processes. Therefore, a single tweet can provide only limited spatiotemporal and evolutionary information, and the generation of disaster evolution chains depends strongly on large-scale data aggregation and relation identification under temporal, spatial and event semantic consistency constraints. When a disaster is small in scale, short in duration, or occurs in an area with low social media activity, key events may be missing and spatiotemporal links may be insufficient, which can affect the completeness and stability of inferred disaster evolution chains.

Second, the consistency between the reconstructed evolutionary chain and post-disaster reports should be interpreted within the broader information environment of disaster communication. During major flood events, social media posts, news reports, public warnings and official summaries may partially share information flows, which means that these sources cannot always be treated as fully independent. In this study, post-disaster reports are used as supplementary references for event-level plausibility analysis rather than as direct inputs for evolutionary chain generation. The reconstruction process itself is based on the tweet-derived KG, with event nodes further linked

back to source tweets for evidence alignment. Future work should further incorporate source provenance tracking and temporally separated information sources to better distinguish real-time social observations from information circulated through media and official channels.

Third, because LLMs are involved in knowledge extraction and result evaluation, model hallucination and potential bias remain unavoidable. To mitigate this issue, this study introduces FO constraints, an evidence-anchored few-shot CoT strategy, and a multi-model cross-evaluation mechanism, together with external validation using a manually annotated gold standard dataset, which can improve the consistency and reliability of knowledge extraction and reconstruction to some extent. However, because LLMs still rely on large-scale pretraining corpora and probabilistic generation mechanisms, their dependence on implicit prior knowledge and language patterns cannot be fully eliminated. Future work should further explore model bias calibration and trustworthy knowledge extraction and reconstruction mechanisms for LLM-driven GeoAI.

Finally, the proposed framework is currently designed mainly for flood disaster scenarios, but its methodological structure is extensible. By reconstructing domain ontologies, adjusting prompt templates, and replacing 3D scene representation rules, the method may be extended to other hazards, such as earthquakes, landslides, debris flows and urban pluvial flooding. However, different disasters vary substantially in triggering mechanisms, evolution patterns and observation characteristics, and direct transfer may reduce knowledge extraction, evolution pathway reconstruction and scene representation performance. Future research will integrate multi-source observations and multi-hazard scenarios to further extend and optimize the framework.

6. Conclusions

This study develops an integrated framework for representing flood spatiotemporal evolution from social media information to knowledge extraction, evolution pathway reconstruction and 3D dynamic representation. By injecting FO constraints into LLM-driven knowledge extraction, fragmented social sensing information is transformed into a structured flood KG. Based on temporal, spatial and event semantic constraints, the main flood disaster evolution pathway is further reconstructed and mapped into a 3D geographic scene, enabling a dynamic representation of flood occurrence, development and impacts.

The experimental results show that the proposed ontology-constrained and evidence-anchored few-shot CoT strategy improves the accuracy, completeness and consistency of flood knowledge extraction, while reducing hallucinated generation. The reconstructed flood evolution pathway shows good stability in evolutionary logic, completeness and spatiotemporal consistency. User evaluation further indicates that the knowledge-guided 3D augmented representation performs better than textual descriptions or basic 3D representation in supporting overall understanding, evolutionary clarity and temporal consistency.

Overall, this work provides a new methodological approach for reconstructing flood disaster evolution pathway from large-scale social media data and communicating complex disaster processes through interpretable 3D geovisualization.

Notes

1. https://www.undrr.org/publication/human-cost-disasters-overview-last-20-years-2000-2019
2. https://youtu.be/CWtj0VZUM6g
3. https://www.bilibili.com/video/BV1zHFKzaEPs/?share_source=copy_web

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Data and codes availability statement

The data and codes that support the findings of this study can be found online at https://doi.org/10.6084/m9.figshare.34025157.

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